TP 5

Serverless Functions

Objectives:

The goal of this TP is to introduce you to Serverless Functions in Microsoft Azure.

We will do the following tasks:

- Develop a Serverless Function.
- Run the Serverless Function locally.
- Call the Serverless Function locally from a python script.

Evaluation:

These TPs have two parts that are evaluated:

1. Experimental part: call the Professor to verify what you did when it tells you to do so.
2. Assignment: an assignment needs to be done at the end of the class. Please send it by email to your TP teacher **before midnight the day of the TP**:
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The TP must be done in groups of 2 to 3 students. The groups must stay the same for all the next TPs.

I- Install the development environment:

In this TP, we will develop a simple REST API for a calculator service using serverless functions. The serverless functions will be developed in python and deployed in Microsoft Azure. We thus need to install the following development environment:

1. Azure Functions Core tools: https://learn.microsoft.com/fr-fr/azure/azure-functions/functions-run-local?WT.mc_id=academic-17441-jabenn&tabs=v4%2Cwindows%2Ccsharp%2Cportal%2Cbash

2. Visual Studio Code: https://code.visualstudio.com/

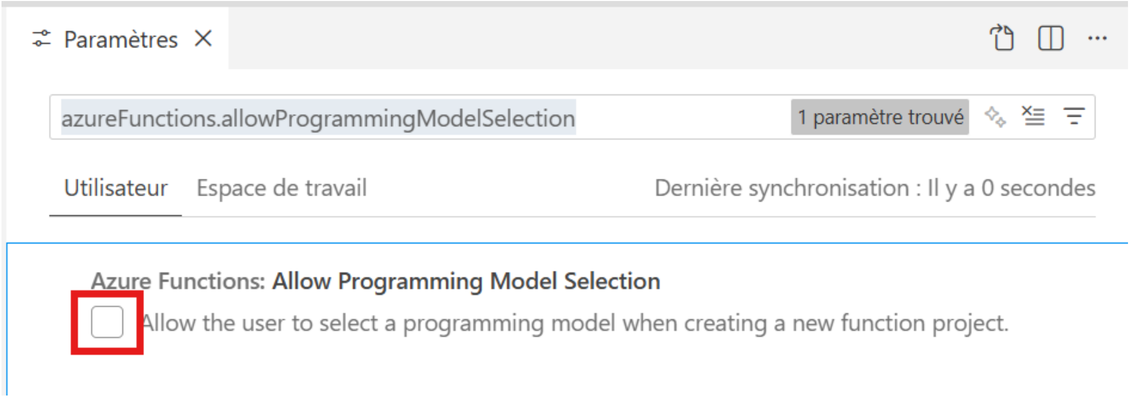
3. Azure Functions extension for Visual Studio Code: https://marketplace.visualstudio.com/items?WT.mc_id=academic-17441-jabenn&itemName=ms-azuretools.vscode-azurefunctions

4. You must have a **Python 3.14** environment installed. For this, you can either install Python 3.14 directly (https://www.python.org/downloads/) or create a 3.14 environment using Miniconda (https://www.anaconda.com/download/success?reg=skipped-miniconda)

5. Open a Powershell terminal in administrator mode, and run the following command to enable the execution of Powershell scripts in your systems: **Set-ExecutionPolicy -ExecutionPolicy RemoteSigned**

II- Create a Serverless Function:

1. Once everything is installed, open **Visual Studio Code**
2. Open the **Settings** of Visual Studio Code by clicking on : **File > Preferences > Settings**
3. Search for the **azureFunctions.allowProgrammingModelSelection** setting and activate it:



azureFunctions.allowProgrammingModelSelection

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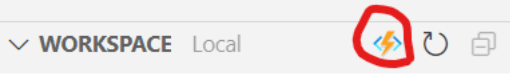
Azure Functions: **Allow Programming Model Selection**

☐ Allow the user to select a programming model when creating a new function project.

4. Click on the Azure icon on the left:

5. Click then on Sign in to Azure... and connect with your Azure UIR account.

6. Click now on the little lightning icon next to **WORKSPACE,** and click on **Create a new Project** to create a new serverless Function:



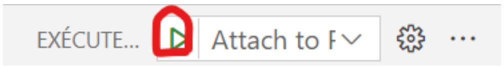
    1. Folder: create a new folder called **calculatorFunction** where the code of the function will be saved
    2. Language: select **Python**
    3. Python Programming Model: Select **Model V1**
    4. Python interpreter: select **Python 3.14.** If it's not found, click on "Manually enter Python interpreter path" and enter the path of the **python.exe** program located in your Python 3.14 installation folder.
    5. Template: **HTTP Trigger**
    6. Function Name: **AddFunction**
    7. Authorization Level: **Function**
    8. How you would like to open your project: **Open in current window**

7. The code of the function will now be generated.
8. Call the professor to show him the code of the created function.

III- Start the function and test it:

1. Click on the **Run and Debug** icon on the left: .

2. Click on the play icon to start the function:



3. If you receive a message regarding an Azure storage account, click on **use local emulator** and then click on **Install Azurite** (if it is not installed) and then click on **Run Emulator**

4. After the function starts, you will see a **url** in the terminal. Open this **url** in your web browser to call the function.

5. Call the professor to show him.

IV- Analyze the default function:

1. The function project contains multiple files. Open the different files, and answer the following questions:
    1. What is the file that contains the code of the function?
    2. What is the file that contains the configuration of the function (e.g., the HTTP methods that you can use to call it, like Get or Post)?
    3. What is the file that contains the name of the python libraries that your function uses?

2. After you look at the code, update the configuration of the function to only accept **Post** requests, and relaunch it.
3. Call the professor to show him.
4. Find a way to call the function with a **Post** request containing the parameter **name** (containing your name) in **JSON** format in its body (hint: you can use a very similar Python script to what you used to query the banana classifier).
5. Call the professor to show him.

V- Implement the Add function:

1. Update the code of the function to receive **two numbers** as parameters in **JSON** format and return the result of their **addition**. Make sure that if the parameters are not provided, or are incorrect (e.g., not numbers) to return an error message with the **HTTP code 400**.
2. Run your function and test it.
3. Call the professor to show him.

VI- Assignment:

Based on what you learned in the TP explain the different steps needed to build an HTTP trigger serverless function, run it locally, and test it. Make sure to explain the different steps you do and use screenshots. Also, make sure you add your answers to the questions in section IV, as well as the code of the script you used to query the function.